Quantum Mechanics, Chapter 1

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Problem 1

(1)

我们取空间中面元 S，以法线方向建立球坐标。

则单位时间内能流的谱密度为

$$\begin{aligned} M(\lambda, T) &= \int_0^{2\pi} \int_0^{\frac{\pi}{2}} B(\lambda, T) \cos\theta (\sin\theta \, \mathrm{d}\theta) \, \mathrm{d}\varphi \\ &= B(\lambda, T) \int_0^{2\pi} \mathrm{d}\varphi \int_0^{\frac{\pi}{2}} \cos\theta \sin\theta \, \mathrm{d}\theta \\ &= \pi B(\lambda, T) = \pi \frac{2hc^2}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1} \\ &\approx 9.93 \times 10^{13} \ \ \mathrm{W \cdot m^{-3}}. \end{aligned}$$

其中 $B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$ 是能流的角谱密度。

(2)

德布罗意关系：

$$\lambda \sim \frac{h}{p} = \frac{h}{mv}.$$

速度

$$v \sim \sqrt{\frac{8k_B T}{\pi m}}.$$

得

$$\lambda \sim \frac{h}{\sqrt{\frac{8k_B T}{\pi m}} \cdot m} = \frac{h}{\sqrt{\frac{8mk_B T}{\pi}}} \approx 3.32 \times 10^{-11} \ \ \mathrm{m}.$$

数密度 n 和钠原子平均距离 d 满足

$$nd^3 \approx 1.$$

得

$$d \approx n^{-\frac{1}{3}} \approx 2.15 \times 10^{-7} \ \ \mathrm{m}.$$

因此

$$d \gg \lambda.$$

没有明显的量子效应。

Problem 2

(1)

不妨取 A 为实数，由归一化条件

$$\begin{aligned} 1 &= \|\psi(x)\|^2 \\ &= \int_R \psi^*(x)\psi(x) \, \mathrm{d}x \\ &= A^2 \int_0^{\infty} x^2 \exp(-2\lambda x) \, \mathrm{d}x \\ &= A^2 \frac{\Gamma(3)}{(2\lambda)^3} \\ &= \frac{A^2}{4\lambda^3}. \end{aligned}$$

因此

$$1 = \frac{A^2}{4\lambda^3}, \quad A = 2\lambda^{\frac{3}{2}}.$$

(2)

首先计算 $\overline{x}$

$$\begin{aligned}
\overline{x} &= \int_{-\infty}^{+\infty} \psi^* x \psi \, \mathrm{d}x \\
&= A^2 \int_0^{\infty} x^3 \exp(-2\lambda x) \, \mathrm{d}x \\
&= A^2 \frac{\Gamma(4)}{(2\lambda)^4} \\
&= 4\lambda^3 \cdot \frac{6}{16\lambda^4} \\
&= \frac{3}{2\lambda}.
\end{aligned}$$

然后计算 $\overline{x^2}$

$$\begin{aligned}
\overline{x^2} &= \int_{-\infty}^{+\infty} \psi^* x^2 \psi \, \mathrm{d}x \\
&= A^2 \int_0^{\infty} x^4 \exp(-2\lambda x) \, \mathrm{d}x \\
&= 4\lambda^3 \cdot \frac{\Gamma(5)}{(2\lambda)^5} \\
&= 4\lambda^3 \cdot \frac{24}{32\lambda^5} \\
&= \frac{3}{\lambda^2}.
\end{aligned}$$

(3)

动量表象下，我们有

$$\begin{aligned}
\varphi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \exp\left(-ip\frac{x}{\hbar}\right) \psi(x) \, \mathrm{d}x \\
&= \frac{2\lambda^{\frac{3}{2}}}{\sqrt{2\pi\hbar}} \int_0^{\infty} x \exp\left(-\left(i\frac{p}{\hbar} + \lambda\right)x\right) \mathrm{d}x \\
&= \frac{2\lambda^{\frac{3}{2}}}{\sqrt{2\pi\hbar}} \frac{1}{\left(\lambda + i\frac{p}{\hbar}\right)^2}.
\end{aligned}$$

对应的概率密度函数为

$$|\varphi(p)|^2 = \frac{2\lambda^3}{\pi\hbar} \frac{1}{\left(\lambda^2 + \frac{p^2}{\hbar^2}\right)^2}.$$

(4)

在动量表象下计算，首先计算 $\overline{p}$

$$\begin{aligned}
\overline{p} &= \int_p \varphi^*(p) p \varphi(p) \, \mathrm{d}p \\
&= \left(\frac{2\lambda^{\frac{3}{2}}}{\sqrt{2\pi\hbar}}\right)^2 \int_{-\infty}^{+\infty} \frac{p}{\left(\lambda^2 + \left(\frac{p}{\hbar}\right)^2\right)^2} \, \mathrm{d}p.
\end{aligned}$$

由于积分号内被积函数是奇函数，

$$\overline{p} = 0$$

。

$$\begin{aligned}\overline{p^2} &= \int_p \varphi^*(p) p^2 \varphi(p)\,\mathrm{d}p \\ &= \frac{2\lambda^3}{\pi\hbar}\int_{-\infty}^{+\infty} \frac{p^2}{\left(\lambda^2+\left(\frac{p}{\hbar}\right)^2\right)^2}\,\mathrm{d}p \\ &= \frac{2\lambda^3}{\pi\hbar}\cdot 2\pi i \sum_{\operatorname{Im} p_i>0} \operatorname{Res}\left(\frac{p^2}{\left(\lambda^2+\left(\frac{p}{\hbar}\right)^2\right)^2}, p_i\right) \\ &= \frac{4\lambda^3 i}{\hbar}\operatorname{Res}\left(\frac{p^2}{\left(\lambda^2+\left(\frac{p}{\hbar}\right)^2\right)^2}, i\hbar\lambda\right) \\ &= \frac{4\lambda^3 i}{\hbar}\cdot\frac{-i\hbar^3}{4\lambda} = \lambda^2\hbar^2.\end{aligned}$$

(5)

验证不确定性关系

$$\sqrt{\overline{(\Delta x)^2}}\cdot\sqrt{\overline{(\Delta p)^2}} \geq \frac{\hbar}{2}.$$

我们分别有

$$\begin{aligned}\sqrt{\overline{(\Delta x)^2}} &= \sqrt{\overline{x^2}-\overline{x}^2} = \frac{\sqrt{3}}{2\lambda}, \\ \sqrt{\overline{(\Delta p)^2}} &= \sqrt{\overline{p^2}-\overline{p}^2} = \lambda\hbar.\end{aligned}$$

则

$$\sqrt{\overline{(\Delta x)^2}}\cdot\sqrt{\overline{(\Delta p)^2}} = \frac{\sqrt{3}\hbar}{2} > \frac{\hbar}{2}.$$

满足不确定性关系。

Problem 3

(1)

动量表象下

$$\begin{aligned}\psi_{p(p)} &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi\hbar}}\exp\left(ip_0\frac{x}{\hbar}\right)\exp\left(-ip\frac{x}{\hbar}\right)\mathrm{d}x \\ &= \frac{1}{2\pi\hbar}\cdot 2\pi\delta\left(\frac{p-p_0}{\hbar}\right) \\ &= \delta(p-p_0).\end{aligned}$$

由于所有可能集中 p_0，故统计涨落 $(\Delta p)^2 = 0$ 。

(2)

我们假设已经知道 x 表象下

$$\hat{x} = x, \quad \hat{p} = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}.$$

我们记 $\varphi(x) \leftrightarrow \psi(p)$ 是各自表象下波函数。

由于力学量在不同表象下是一致的，我们有

$$\int_x \varphi^*(x) x \varphi(x)\,\mathrm{d}x = \int_p \psi^*(p)\hat{x}\psi(p)\,\mathrm{d}p$$

和

$$\int_x \varphi^*(x)(-i\hbar)\frac{\mathrm{d}}{\mathrm{d}x}\varphi(x)\,\mathrm{d}x = \int_p \psi^*(p)\hat{p}\psi(p)\,\mathrm{d}p$$

首先推导 $\hat{p}$ 在动量表象下的表示

$$\begin{aligned}\int_x \varphi^*(x)\hat{p}\varphi(x)\,\mathrm{d}x &= \int_x \varphi^*(x)(-i\hbar)\frac{\mathrm{d}}{\mathrm{d}x}\varphi(x)\,\mathrm{d}x \\ &= \int_x \left(\frac{1}{\sqrt{2\pi\hbar}}\int_p \exp\left(-ix\frac{p}{\hbar}\right)\psi^*(p)\,\mathrm{d}p\right)(-i\hbar)\frac{\mathrm{d}}{\mathrm{d}x}\varphi(x)\,\mathrm{d}x \\ &= \int_p \frac{\psi^*(p)}{\sqrt{2\pi\hbar}}\int_x (-i\hbar)\exp\left(-ix\frac{p}{\hbar}\right)\frac{\mathrm{d}}{\mathrm{d}x}\varphi(x)\,\mathrm{d}x\,\mathrm{d}p. \\ &\overset{\text{由分部积分}}{=} \int_p \frac{\psi^*(p)}{\sqrt{2\pi\hbar}}\left((-i\hbar)\left[\exp\left(-ix\frac{p}{\hbar}\right)\varphi(x)\right]_{-\infty}^{+\infty} - (-i\hbar)\left(-i\frac{p}{\hbar}\right)\int_x \exp\left(-ix\frac{p}{\hbar}\right)\varphi(x)\,\mathrm{d}x\right)\mathrm{d}p. \\ &\overset{\text{束缚态情形下边界项消失}}{=} \int_p \psi^*(p)\frac{1}{\sqrt{2\pi\hbar}}(-i\hbar)\left(-i\frac{p}{\hbar}\right)\left(\int_x \exp\left(-ix\frac{p}{\hbar}\right)\varphi(x)\,\mathrm{d}x\right)\mathrm{d}p \\ &= \int_p \psi^*(p)p\psi(p)\,\mathrm{d}p.\end{aligned}$$

因此动量表象下

$$\hat{p} = p.$$

下面推导 $\hat{x}$ 在动量表象下的表示，同理：

$$\begin{aligned}&\int_x \varphi^*(x)x\varphi(x)\,\mathrm{d}x \\ &= \int_x \varphi^*(x)x\frac{1}{\sqrt{2\pi\hbar}}\left(\int_p \psi(p)\exp\left(ip\frac{x}{\hbar}\right)\mathrm{d}p\right)\mathrm{d}x \\ &= \int_x \frac{\varphi^*(x)}{\sqrt{2\pi\hbar}}\int_p \psi(p)x\exp\left(ip\frac{x}{\hbar}\right)\mathrm{d}p\,\mathrm{d}x.\end{aligned}$$

注意

$$x\exp\left(ip\frac{x}{\hbar}\right) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}p}\exp\left(ip\frac{x}{\hbar}\right).$$

于是上上式化作

$$\begin{aligned}&\int_x \frac{\varphi^*(x)}{\sqrt{2\pi\hbar}}\int_p \psi(p)(-i\hbar)\frac{\mathrm{d}\left(\exp\left(ip\frac{x}{\hbar}\right)\right)}{\mathrm{d}p}\,\mathrm{d}p\,\mathrm{d}x \\ &= \int_x \frac{\varphi^*(x)(-i\hbar)}{\sqrt{2\pi\hbar}}\left(\left[\psi(p)\exp\left(ip\frac{x}{\hbar}\right)\right]_{-\infty}^{+\infty} - \int_p \exp\left(ip\frac{x}{\hbar}\right)\frac{\mathrm{d}\psi(p)}{\mathrm{d}p}\,\mathrm{d}p\right)\mathrm{d}x. \\ &\overset{\text{束缚态下边界项消失}}{=} \int_x \varphi^*(x)\frac{1}{\sqrt{2\pi\hbar}}\left(\int_p \exp\left(ip\frac{x}{\hbar}\right)\left(i\hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)\psi(p)\,\mathrm{d}p\right)\mathrm{d}x. \\ &\overset{\text{交换积分次序}}{=} \int_p \left(\int_x \frac{\varphi^*(x)\exp\left(ip\frac{x}{\hbar}\right)}{\sqrt{2\pi\hbar}}\,\mathrm{d}x\right)\left(i\hbar\frac{\mathrm{d}}{\mathrm{d}p}\right)\psi(p)\,\mathrm{d}p \\ &= \int_p \psi^*(p)i\hbar\frac{\mathrm{d}}{\mathrm{d}p}\psi(p)\,\mathrm{d}p.\end{aligned}$$

由此，$\hat{x}$ 在动量表象下就是 $i\hbar\frac{\mathrm{d}}{\mathrm{d}p}$

(3)

自由粒子的哈密顿量为

$$\hat{H} = \frac{\hat{p}^2}{2m}.$$

由上一问，动量表象下 $\hat{p} = p$，因此薛定谔方程直接变为

$$
\begin{aligned}
i\hbar\frac{\partial\varphi(p,t)}{\partial t} &= \hat{H}\varphi(p,t) \\
&= \frac{\hat{p}^2}{2m}\varphi(p,t) \\
&= \frac{p^2}{2m}\varphi(p,t).
\end{aligned}
$$

所以在动量表象下，自由粒子的波函数满足

$$
i\hbar\frac{\partial\varphi(p,t)}{\partial t} = \frac{p^2}{2m}\varphi(p,t).
$$

Problem 4

我们有

$$
\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \frac{1}{\sqrt{34}}\begin{pmatrix}3 & 5\\ 5 & -3\end{pmatrix}\begin{pmatrix}\chi_1\\ \chi_2\end{pmatrix}
$$

和

$$
\begin{pmatrix}\chi_1\\ \chi_2\end{pmatrix} = \frac{1}{\sqrt{34}}\begin{pmatrix}3 & 5\\ 5 & -3\end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}.
$$

测量 $\hat{A}$ 后得到 α_1，量子态为 ψ_1 有

$$
\psi_1 = \frac{1}{\sqrt{34}}(3\chi_1 + 5\chi_2).
$$

$$
P(\chi_1) = \left(\frac{3}{\sqrt{34}}\right)^2 = \frac{9}{34}.
$$

$$
P(\chi_2) = \left(\frac{5}{\sqrt{34}}\right)^2 = \frac{25}{34}.
$$

测得 $\hat{B}$ 时，若为 β_1

$$
\chi_1 = \frac{1}{\sqrt{34}}(3\psi_1 + 5\psi_2).
$$

$$
P(\psi_1|\chi_1) = \left(\frac{3}{\sqrt{34}}\right)^2 = \frac{9}{34}
$$

若测得 β_2

$$
\chi_2 = \frac{1}{\sqrt{34}}(5\psi_1 - 3\psi_2).
$$

$$
P(\psi_1|\chi_2) = \left(\frac{5}{\sqrt{34}}\right)^2 = \frac{25}{34}.
$$

则

$$
P(\psi_1) = P(\psi_1|\chi_1)P(\chi_1) + P(\psi_1|\chi_2)P(\chi_2) = \frac{353}{578}.
$$

Problem 5

(1)

由薛定谔方程

$$
i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi.
$$

取共轭

$$
-i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*.
$$

现在计算

$$\begin{aligned}\frac{\partial}{\partial t}\|\psi\|^2 &= \psi\frac{\partial\psi^*}{\partial t}+\psi^*\frac{\partial\psi}{\partial t}.\\ &= \frac{i\psi}{\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi^*+V\psi^*\right)+\frac{-i\psi^*}{\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi+V\psi\right)\\ &= \frac{\hbar}{2mi}(\psi\nabla^2\psi^*-\psi^*\nabla^2\psi).\end{aligned}$$

由恒等式

$$\nabla\cdot(f\boldsymbol{A})=(\nabla f)\cdot\boldsymbol{A}+f(\nabla\cdot\boldsymbol{A}).$$

代入 $f=\psi^*,\boldsymbol{A}=\nabla\psi$

$$\nabla\cdot(\psi^*\nabla\psi)=(\nabla\psi^*)\cdot\nabla\psi+\psi^*\nabla^2\psi.$$

得

$$\psi^*\nabla^2\psi=\nabla\cdot(\psi^*\nabla\psi)-(\nabla\psi^*)\cdot(\nabla\psi).$$

同理

$$\psi\nabla^2\psi^*=\nabla\cdot(\psi\nabla\psi^*)-(\nabla\psi)\cdot(\nabla\psi^*).$$

将上二式代入 $\frac{\partial}{\partial t}\|\psi\|^2$，得到

$$\begin{aligned}\frac{\partial}{\partial t}\|\psi\|^2 &= \frac{\hbar}{2mi}(\psi\nabla^2\psi^*-\psi^*\nabla^2\psi).\\ &= -\nabla\cdot\left(\frac{\hbar}{2mi}(\psi^*\nabla\psi-\psi\nabla\psi^*)\right)=-\nabla\cdot\boldsymbol{j}.\end{aligned}$$

其中 $\boldsymbol{j}$ 为

$$\boldsymbol{j}=\frac{\hbar}{2mi}(\psi^*\nabla\psi-\psi\nabla\psi^*).$$

最终得到微分形式

$$\frac{\partial}{\partial t}\|\psi\|^2+\nabla\cdot\boldsymbol{j}=0.$$

从微分形式积分

$$\int_V\frac{\partial}{\partial t}\|\psi\|^2\,\mathrm{d}V=\int_V-\nabla\cdot\boldsymbol{j}\,\mathrm{d}V.$$

左式化成

$$\frac{\partial}{\partial t}\int_V\|\psi\|^2\,\mathrm{d}V.$$

右式由 Gauss 定理化成

$$-\int_V\nabla\cdot\boldsymbol{j}\,\mathrm{d}V=-\int_{\partial V}\boldsymbol{j}\cdot\mathrm{d}\boldsymbol{S}.$$

$V\to\infty$ 时，由于边界 ∂V 上 ψ 和 ψ^* 都趋于零，后式

$$\int_{\partial V}\boldsymbol{j}\cdot\mathrm{d}\boldsymbol{S}\to 0.$$

则得到积分形式

$$\frac{\partial}{\partial t}\int_V\|\psi\|^2\,\mathrm{d}V=0.$$

(2)

重复上面步骤，但加入 iV_2，得到

$$\begin{aligned}\frac{\partial}{\partial t}\|\psi\|^2+\nabla\cdot\boldsymbol{j} &= \frac{i\psi}{\hbar}(-iV_2\psi^*)+\frac{-i\psi^*}{\hbar}(iV_2\psi)\\ &= \frac{2}{\hbar}V_2\|\psi\|^2.\end{aligned}$$

对微分形式求积分就得

$$\frac{\partial}{\partial t}\int_V\|\psi\|^2\,\mathrm{d}V=-\frac{\hbar}{2im}\int_{\partial V}(\psi^*\nabla\psi-\psi\nabla\psi^*)\cdot\mathrm{d}\boldsymbol{S}+\frac{2}{\hbar}\int_V V_2\|\psi\|^2\,\mathrm{d}V.$$

可见，概率在有界空间内并不总是守恒